# WP1 Solutions

3. $P(-1,3)$, $Q(1,5)$, $R(-3,7)$

$\overrightarrow{QP} = \langle -1-1, 3-5 \rangle$
$= \langle -2, -2 \rangle$
$\overrightarrow{QP} = -2\hat{\imath} - 2\hat{\jmath}$

7. $P(-1,3)$, $Q(1,5)$, $R(-3,7)$

$2\overrightarrow{PQ} - 2\overrightarrow{PR} = 2\langle 1+1, 5-3 \rangle - 2\langle -3+1, 7-3 \rangle$
$= 2\langle 2,2 \rangle - 2\langle -2,4 \rangle$
$= \langle 4,4 \rangle - \langle -4,8 \rangle$
$= \langle 8,-4 \rangle$
$2\overrightarrow{PQ} - 2\overrightarrow{PR} = 8\hat{\imath} - 4\hat{\jmath}$

13. $P(1,0)$, $Q$ on $y$-axis, magnitude $= \sqrt{5}$

$\sqrt{5} = \sqrt{(1-0)^2 + (0-y)^2}$
$\sqrt{5} = \sqrt{1+y^2}$
$5 = 1 + y^2$
$4 = y^2$
$y = \pm 2$
since this is above $P$, need $y > 0$.
So $Q(0,2)$

15. $a = 2\hat{\imath} + \hat{\jmath}$, $b = \hat{\imath} + 3\hat{\jmath}$

a) $a+b = (2\hat{\imath}+\hat{\jmath}) + (\hat{\imath}+3\hat{\jmath})$
$= 3\hat{\imath} + 4\hat{\jmath}$ and $\langle 3,4 \rangle$

b) $a-b = (2\hat{\imath}+\hat{\jmath}) - (\hat{\imath}+3\hat{\jmath})$
$= \hat{\imath} - 2\hat{\jmath}$ and $\langle 1,-2 \rangle$

c) $\|a\| + \|b\| = \sqrt{2^2+1^2} + \sqrt{1^2+3^2}$
$= \sqrt{5} + \sqrt{10}$
$\|a+b\| = \sqrt{3^2+4^2}$
$= \sqrt{25}$
$= 5$

so $\|a\| + \|b\| \leq \|a+b\|$

d) $2a = 2(2\hat{\imath}+\hat{\jmath})$
$= 4\hat{\imath} + 2\hat{\jmath}$
and $\langle 4,2 \rangle$

$-b = -(\hat{\imath}+3\hat{\jmath})$
$= -\hat{\imath} - 3\hat{\jmath}$
and $\langle -1,-3 \rangle$

$2a - b = (4\hat{\imath}+2\hat{\jmath}) + (-\hat{\imath}-3\hat{\jmath})$
$= 3\hat{\imath} - \hat{\jmath}$ and $\langle 3,-1 \rangle$

25. $\|v\| = 7$, $u = \langle 3,4 \rangle$

$\|u\| = \sqrt{3^2+4^2}$
$= \sqrt{25}$
$= 5$
$v = \frac{7}{5}\langle 3,4 \rangle$
$= \left\langle \frac{21}{5}, \frac{28}{5} \right\rangle$

45. $v_0 = 70\text{mph}, \theta_0 = 30^\circ$

$V = \langle v_0 \cos\theta, v_0 \sin\theta \rangle$
$= \langle 70\cos 30^\circ, 70 \sin 30^\circ \rangle$
$\langle 35\sqrt{3}, 35 \rangle$
$V \approx \langle 60.62, 35 \rangle$

81. $\overrightarrow{PQ} = \langle 7, -1, 3 \rangle, P(-2,3,5)$

$\langle 7,-1,3 \rangle = \langle x+2, y-3, z-5 \rangle$
$x+2=7 \quad y-3=-1 \quad z-5=3$
$x=5 \quad y=2 \quad z=8$

so $Q(5,2,8)$

99. $V = \langle 7,-1,3 \rangle, \|u\| = 10$

$\|v\| = \sqrt{7^2+(-1)^2+3^2}$
$= \sqrt{49+1+9}$
$= \sqrt{59}$

$u = \frac{10}{\sqrt{59}} \langle 7,-1,3 \rangle$
$= \left\langle \frac{70}{\sqrt{59}}, \frac{-10}{\sqrt{59}}, \frac{30}{\sqrt{59}} \right\rangle$
$u = \left\langle \frac{70\sqrt{59}}{59}, \frac{-10\sqrt{59}}{59}, \frac{30\sqrt{59}}{59} \right\rangle$

77. $P(3,0,2)$ and $Q(-1,-1,4)$

a) $\overrightarrow{PQ} = \langle -1-3, -1-0, 4-2 \rangle$
$= \langle -4, -1, 2 \rangle$

b) $\overrightarrow{PQ} = -4\hat{i} - \hat{j} + 2\hat{k}$

89. $u = \langle 2\cos t, -2\sin t, 3 \rangle, v = \langle 0,0,3 \rangle$

$\|u-v\| = \sqrt{(2\cos t - 0)^2 + (-2\sin t - 0)^2 + (3-3)^2}$
$= \sqrt{4\cos^2 t + 4\sin^2 t}$
$= 2.$

$\|-2u\| = \sqrt{(-4\cos t)^2 + (4\sin t)^2 + (-6)^2}$
$= \sqrt{16\cos^2 t + 16\sin^2 t + 36}$
$= \sqrt{52}$
$= 2\sqrt{13}$

115. $F_1 = \langle 10,6,3 \rangle, F_2 = \langle 0,4,9 \rangle, F_3 = \langle 10,-3,-9 \rangle$

$F_1 + F_2 + F_3 = \langle 10+0+10, 6+4-3, 3+9-9 \rangle$
$= \langle 20, 7, 3 \rangle$

so $F_4 = \langle -20, -7, -3 \rangle$